🤌 XAI: Understanding Course Foundations and Motivations

FAMAF - UNC

November 5, 2025

eXplainable
Artificial Intelligence

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

Agenda ---

- Course Goals & Logistics
- Model Understanding: Use Cases
- Inherently Interpretable Models vs. Post-hoc Explanations
- Defining and Understanding Interpretability

🧉 Goals of this Course ---

Learn and improve upon the state-of-the-art literature on ML interpretability and explainability:

- Understand where, when, and why is interpretability/explainability needed
- Read, present, and discuss research papers
- Formulate, optimize, and evaluate algorithms
- Implement state-of-the-art algorithms; Do research!
- Understand, critique, and redefine literature

EMERGING FIELD!!

🧉 Course Overview ---

1. **Foundations and Human Factors** - definitions, taxonomies, and cognitive aspects
2. **Interpretability and Explainability Methods** - inherently interpretable models and post-hoc explanations
3. **Advanced Techniques and Evaluation** - attention-based explanations and quality metrics
4. **Interpretability of Large Models** - mechanistic interpretability and LLM understanding
5. **Emerging Frontiers of XAI** - automated circuit discovery and multimodal techniques

🧉 Who should take this course? ---

- Course particularly tailored to students interested in **research** 🧪 on interpretability/explainability
  - **Not a surface level course!** 🔥
  - **Not just applications!** 🔥
- Goal is to push you to question existing work and make new contributions to the field

🧉 Class Format ---

- Course comprises of lectures, guests, and student presentations
- Each lecture will cover: at least 2 papers 📄 📄
- Students are **expected** to "at least" skim through the papers beforehand 🥱
- Students will divide into groups

Each breakout group is expected to come up with:

- A list of 2 to 3 weaknesses of each of the works discussed
- Strategies for addressing those weaknesses

🧉 Research Projects - AKA Final Exam ---

Requirements

- Short research paper (max 4 pages)
- Teams of 2 students
- Target: Local conference submission

Assessment

- Ongoing feedback from entire course
- Course approved when paper is submitted

🧉 Background ---

Understanding:

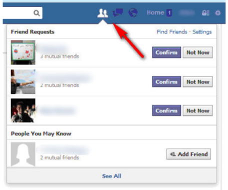
- Linear algebra
- Probability
- Algorithms
- Machine learning
- Programming in Python, Numpy, Sklearn

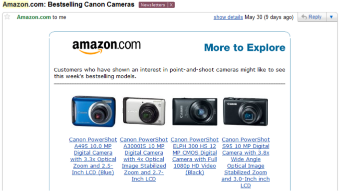
Familiarity with:

- Statistics
- Optimization

🧉 Motivation ---

Machine Learning is EVERYWHERE!!





(Lipton 2016)

🧉 Motivation: Why Model Understanding? ---

Example: Image classification model

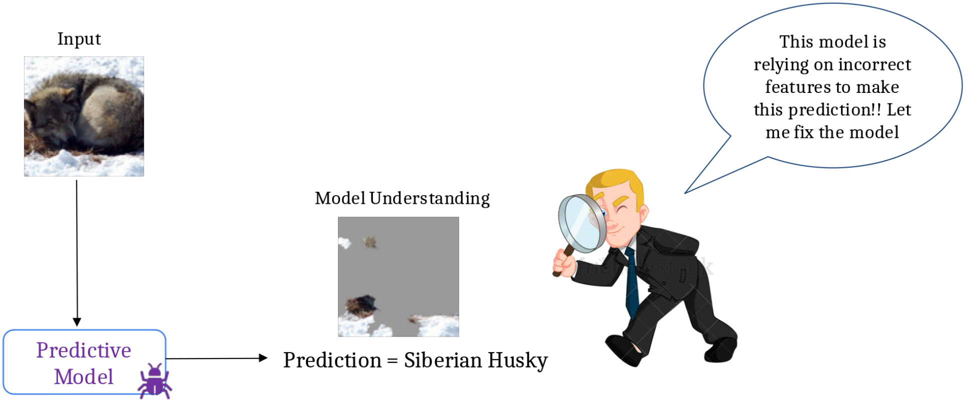
Input

Predictive
Model

Prediction = Siberian Husky

🧉 Motivation: Why Model Understanding? ---

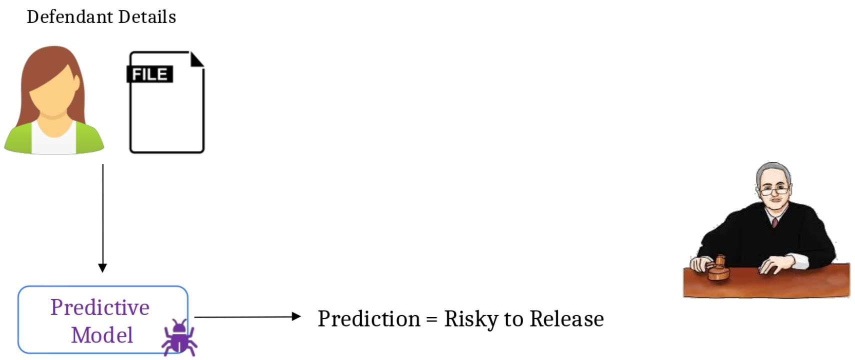
Example: Image classification model



Model understanding facilitates debugging

🧉 Motivation: Why Model Understanding? ---

Example: Criminal justice risk assessment



(Lipton 2016)

🧉 Motivation: Why Model Understanding? ---

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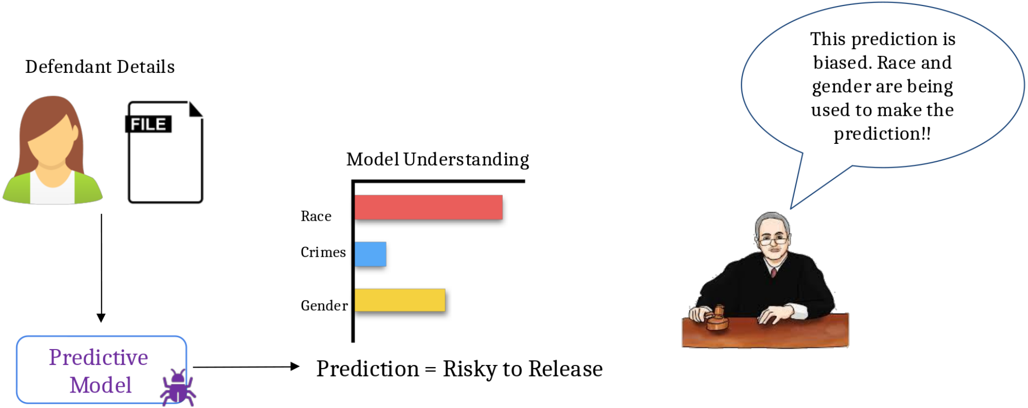
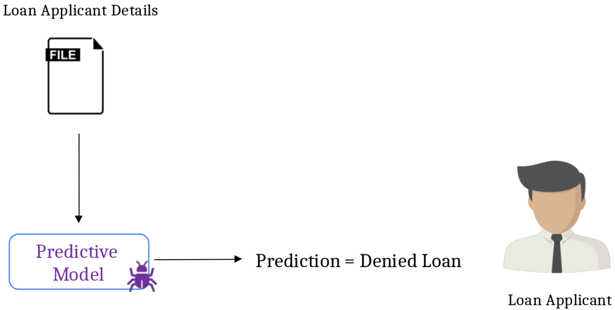


Model understanding facilitates bias detection

(Lipton 2016)

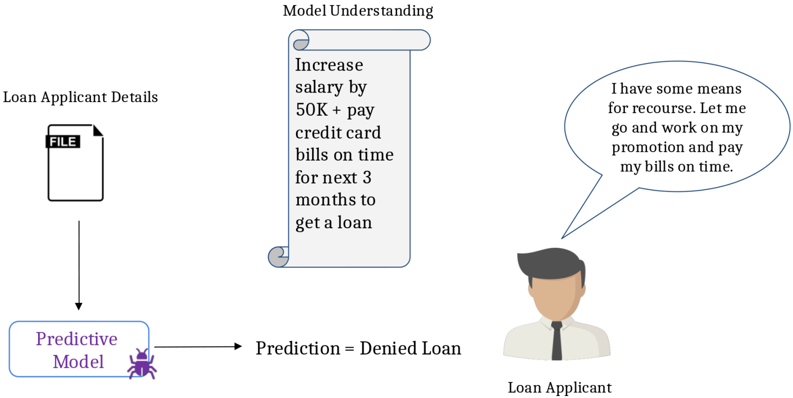
Motivation: Why Model Understanding? ---

Example: Loan application system



🧉 Motivation: Why Model Understanding? ---

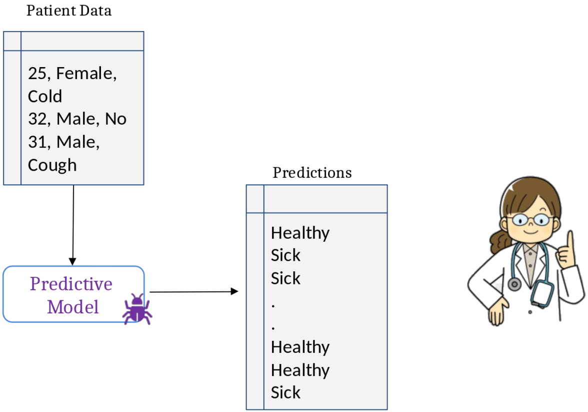
Example: Loan application system



Model understanding helps provide recourse to individuals who are adversely affected by model predictions

🧉 Motivation: Why Model Understanding? ---

Example: Medical diagnosis system



🧉 Motivation: Why Model Understanding? ---

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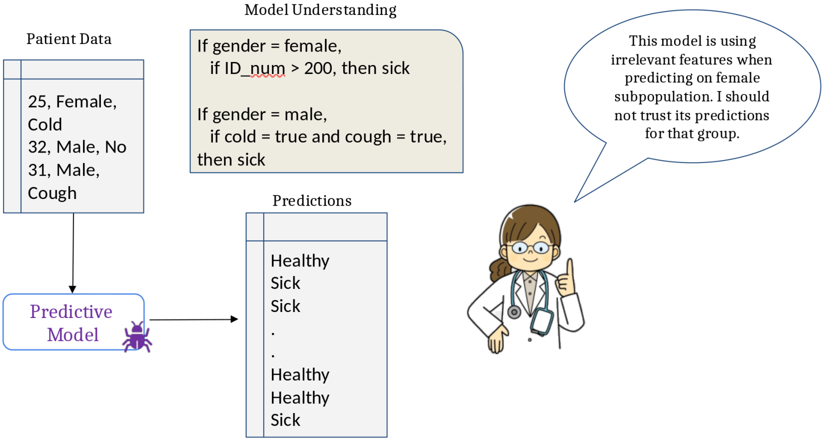

Model understanding helps assess when to trust predictions

🧉 Motivation: Why Model Understanding? ---

Summary of Use Cases

Utility

- Debugging
- Bias Detection
- Recourse
- If and when to trust model predictions
- Vet models to assess suitability for deployment

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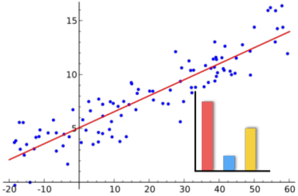
Stakeholders

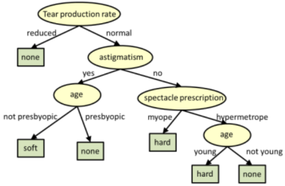
- End users (e.g., loan applicants)
- Decision makers (e.g., doctors, judges)
- Regulatory agencies (e.g., FDA, European commission)
- Researchers and engineers

🧉 Achieving Model Understanding ---

Take 1: Build inherently interpretable predictive models



Linear regression



Decision trees

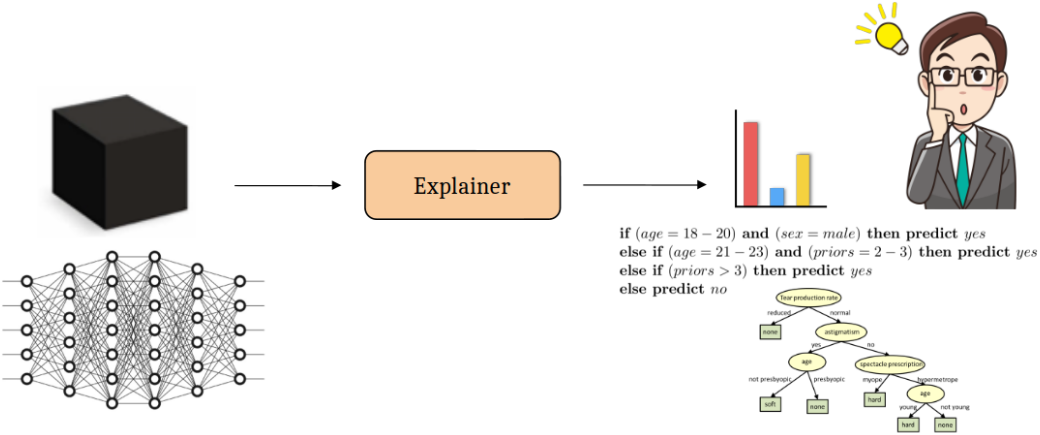
if ($age = 18-20$) **and** ($sex = male$) **then predict** *yes*
else if ($age = 21-23$) **and** ($priors = 2-3$) **then predict** *yes*
else if ($priors > 3$) **then predict** *yes*
else predict *no*

Rule-based models

(Letham et al. 2015; Lakkaraju, Bach, and Leskovec 2016)

🧉 Achieving Model Understanding ---

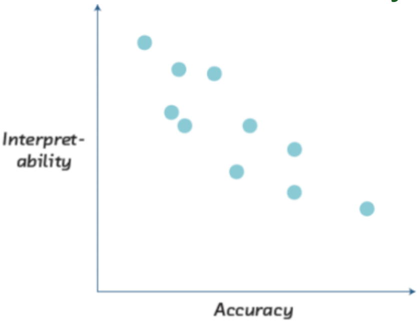
Take 2: Explain pre-built models in a post-hoc manner

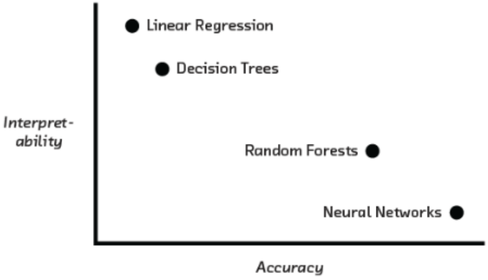


(Ribeiro, Singh, and Guestrin 2016, 2018; Lakkaraju and Lage 2019)

🧉 Inherently Interpretable Models vs. Post hoc Explanations ---

Accuracy-Interpretability Trade-offs



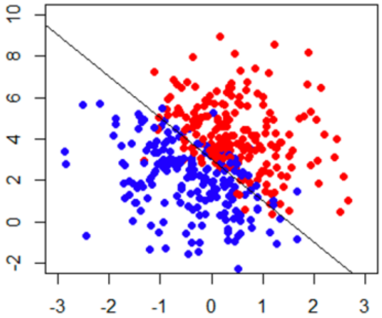


In certain settings, accuracy-interpretability trade offs may exist.

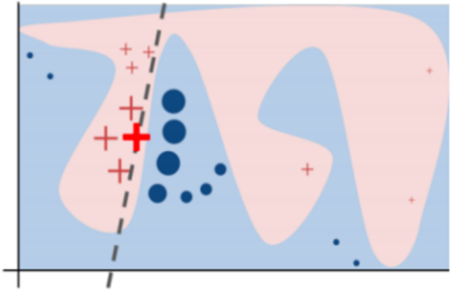
(Cireșan et. al. 2012, Caruana et. al. 2006, Frosst et. al. 2017, Stewart 2020)

🧉 Inherently Interpretable Models vs. Post hoc Explanations ---

Example scenarios: Simple vs complex boundaries



can build interpretable + accurate models



complex models might achieve higher accuracy

In certain settings, accuracy-interpretability trade offs may exist.

🧉 Inherently Interpretable Models vs. Post hoc Explanations ---

Sometimes, you don't have enough data to build your model from scratch. And, all you have is a (proprietary) black box!

(Ribeiro, Singh, and Guestrin 2016)

🧉 Inherently Interpretable Models vs. Post hoc Explanations ---

Recommendation

If you can build an interpretable model which is also adequately accurate for your setting,

DO IT!

Otherwise, **post hoc explanations** *come to the rescue!*

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Let's get into some details!

🧉 Next Up! ---

- Define and evaluate interpretability somewhat! 🤌
- Taxonomy of interpretability evaluation
- Taxonomy of interpretability based on applications/tasks
- Taxonomy of interpretability based on methods

🧉 Defining and Understanding Interpretability: Motivation for Interpretability ---

- ML systems are being deployed in complex **high-stakes settings**
- Accuracy alone is no longer enough
- **Auxiliary criteria are important:**
    - Safety
    - Nondiscrimination
    - Right to explanation

🧉 Motivation for Interpretability (cont.) ---

- Auxiliary criteria are often **hard to quantify** (completely):
  - E.g.: Impossible to enumerate all scenarios violating safety of an autonomous car

Fallback option: *Interpretability*

If the system can explain its reasoning, we can verify if that reasoning is sound w.r.t. auxiliary criteria

Prior Work: Defining and Measuring Interpretability ---

Bad News

Little consensus on what interpretability is and how to evaluate it.

Interpretability evaluation typically falls into:

1. **Evaluate in the context of an application**
   - If a system is useful in a practical application or a simplified version, it must be interpretable
2. **Evaluate via a quantifiable proxy**
   - Claim some model class is interpretable and present algorithms to optimize within that class
   - E.g. rule lists

"You will know it when you see it!"

🧉 Lack of Rigor? ---

Yes and No

Previous notions are reasonable

- **However:**
  - Are all models in all "interpretable" model classes equally interpretable?
    - Model sparsity allows for comparison
  - How to compare a linear model with a decision tree?
  - Do all applications have same interpretability needs?

Important to formalize these notions!!!

🧉 What is Interpretability? ---

Definition

Ability to explain or to present in understandable terms to a human

No clear answers in psychology to:

- What constitutes an explanation?
- What makes some explanations better than the others?
- When are explanations sought?

🧉 When and Why Interpretability? ---

Not all ML systems require interpretability

- E.g., ad servers, postal code sorting
- No human intervention
- **No explanation needed because:**
  - No consequences for unacceptable results
  - Problem is well studied and validated well in real-world applications $\rightarrow$ trust system's decision

When do we need explanation then?

🧉 When and Why Interpretability? ---

- **Incompleteness in problem formalization**
  - Hinders optimization and evaluation

- **Incompleteness $\neq$ Uncertainty**
  - Uncertainty can be quantified
  - E.g., trying to learn from a small dataset (uncertainty)

🧉 Incompleteness: Illustrative Examples ---

Scientific Knowledge

- E.g., understanding the characteristics of a large dataset
- Goal is abstract

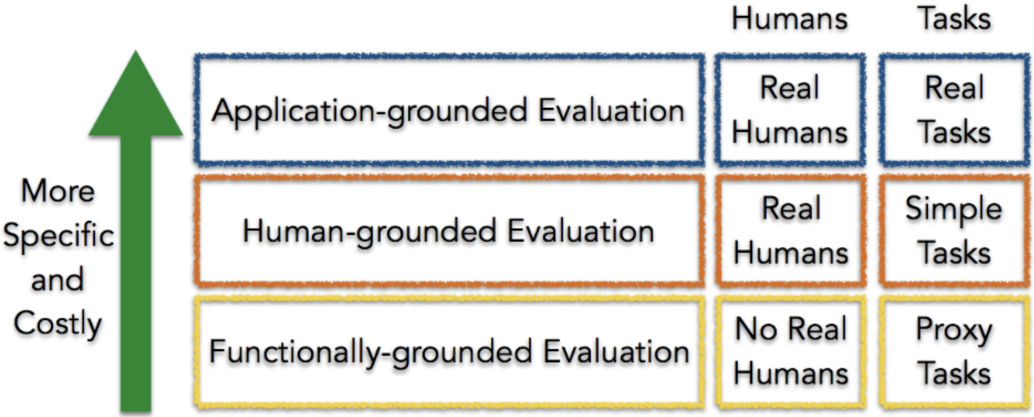
Safety

- End to end system is never completely testable
- Not possible to check all possible inputs

Ethics

- Guard against certain kinds of discrimination which are too abstract to be encoded
- No idea about the nature of discrimination beforehand

🧉 Taxonomy of Interpretability Evaluation ---



Claim of the research should match the type of the evaluation!

🧉 Application-grounded evaluation ---

- Real humans (domain experts), real tasks
- Domain experts experiment with **exact application task**
- Domain experts experiment with a **simpler or partial task**
    - Shorten experiment time
    - Increases number of potential subjects
- Typical in *Human-Computer Interaction* (HCI) and visualization communities

🧉 Human-grounded evaluation ---

Real humans, simplified tasks

- Can be completed with lay humans
- Larger pool, less expensive

Potential experiments:

- Pairwise comparisons
- Simulate the model output
- What changes should be made to input to change the output?

🧉 Functionally-grounded evaluation ---

No humans, just proxies

- Appropriate for a class of models already validated (E.g., decision trees)
- A method is not yet mature
- Human subject experiments are unethical
- What proxies to use?

Potential experiments:

- Complexity (of a decision tree) compared to other models of the same (similar) class
  - How many levels? How many rules?

🧉 Open Problems: Design Issues ---

- What proxies are best for what real world applications? 🤌
- What factors to consider when designing simpler tasks in place of real world tasks? 🤌

🧉 Taxonomy based on applications/tasks ---

Global vs. Local

- High level patterns vs. specific decisions

Degree of Incompleteness

- What part of the problem is incomplete? How incomplete is it?
- Incomplete inputs or constraints or costs?

Time Constraints

- How much time can the user spend to understand explanation?

🧉 Taxonomy based on applications/tasks (cont.) ---

Nature of User Expertise

- How experienced is end user?
- Experience affects how users process information
- E.g., domain experts can handle detailed, complex explanations compared to opaque, smaller ones

Note:

These taxonomies are constructed based on intuition and are not data or evidence driven. They must be treated as hypotheses.

🧉 Taxonomy based on methods ---

Basic units of explanation:

- Raw features? E.g., pixel values
- Semantically meaningful? E.g., objects in an image
- Prototypes?

Number of basic units of explanation:

- How many does the explanation contain?
- How do various types of basic units interact?
- E.g., prototype vs. feature

🧉 Taxonomy based on methods (cont.) ---

Level of compositionality:

- Are the basic units organized in a structured way?
- How do the basic units compose to form higher order units?

Interactions between basic units:

- Combined in linear or non-linear ways?
- Are some combinations easier to understand?

Uncertainty:

- What kind of uncertainty is captured by the methods?
- How easy is it for humans to process uncertainty?

🧉 Relevant Conferences to Explore ---

ML/AI Venues:

- ICML, NeurIPS, ICLR
- UAI, AISTATS, KDD, AAAI

Ethics/HCI Venues:

- FAccT, AIES
- CHI, CSCW, HCOMP

🧉 Closing ---

- Say hi to your neighbors! Introduce yourselves!
- What topics are you most excited about learning as part of this course?
- Are you convinced that model interpretability/explainability is important?
- Do you think we can really interpret/explain models (correctly)?
- What is your take on inherently interpretable models vs. post hoc explanations? Would you favor one over the other? Why?

🧉 References --- |

Lakkaraju, Himabindu, Stephen H Bach, and Jure Leskovec. 2016. "Interpretable Decision Sets: A Joint Framework for Description and Prediction." In *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, 1675–84.

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